

Opening the Black Box

Jesus D. Jimenez Ballestas · Group 25
· IE7374 Generative AI

Locating, ablating, and editing
the mechanisms of a small
language model

The problem

- A language model is usually a black box: prompt in, text out
- When it states a **wrong fact**, the conventional fixes are retraining or fine-tuning — both change the entire network to correct one thing
- **This project:** treat EleutherAI's **Pythia-160m** as a system with parts
 - Find the parts responsible for three specific behaviors
 - Use that to **change what the model writes** by editing *one weight matrix*
 - Nothing trained, nothing fine-tuned — one deliberate rank-one update

Four research questions

- **RQ1** — which attention heads and MLPs *causally* drive subject–verb agreement, factual recall, and induction?
- **RQ2** — when we ablate them, does performance drop **selectively**?
- **RQ3 (generative core)** — can that localization be used to edit the model so it **generates** the new fact, with efficacy, generalization, specificity and fluency?
- **RQ4** — using Pythia’s 154 training checkpoints, **when** does each behavior appear?

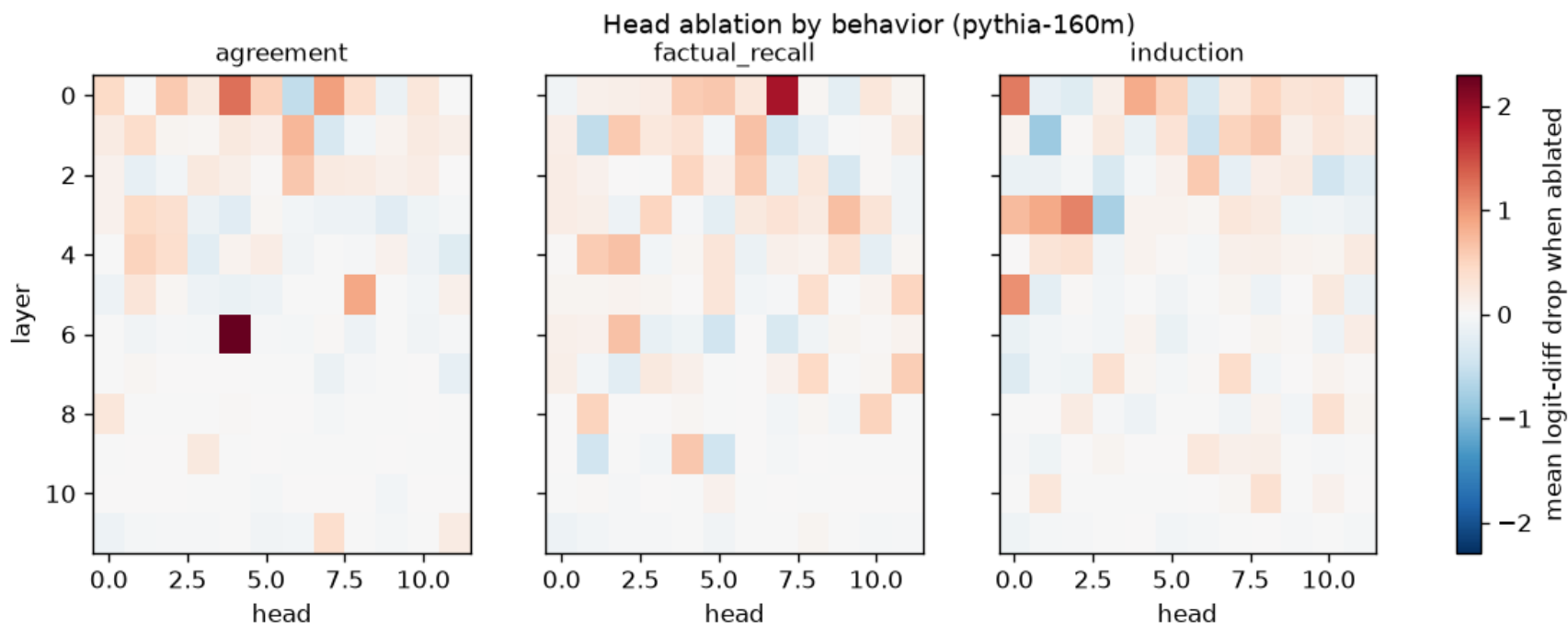
Data: minimal pairs, hand-built

Suite	Prompts	Example	Correct	Incorrect
Agreement	20	The keys on the table	are	is
Factual recall	12	The capital of France is	Paris	London
Induction	16	AliceBob Alice	Bob	Carol

Method: measure → localize → edit

- **Measure** — logit difference: $\text{logit}(\text{correct}) - \text{logit}(\text{incorrect})$, one forward pass
- **Localize**
 - logit lens — where the answer forms (observational)
 - zero-ablation of every head and MLP — causal contribution
 - causal tracing — corrupt the subject, restore layer by layer
- **Edit (ROME-style)** — optimize a delta at the subject's last token, then a closed-form rank-one update to that MLP's output matrix
- **Score the generated text** — efficacy ·

Result 1 — ablating every head, per behavior



One head owns subject–verb agreement

Behavior	Top head	Own behavior	Other behaviors	Selectivity
Agreement	L6.H4	+2.29	−0.02	1.00
Induction	L0.H0	+1.19	+0.17	0.75
Factual recall	L0.H7	+1.89	+0.60	0.52

Result 2 — the edit changes what it writes

Pythia-160m · Mount Everest: Nepal → Canada · one weight matrix differs

before — *Mount Everest is located in the country of Nepal. The mountain is located in the Himalayas, and is the highest mountain in the world...*

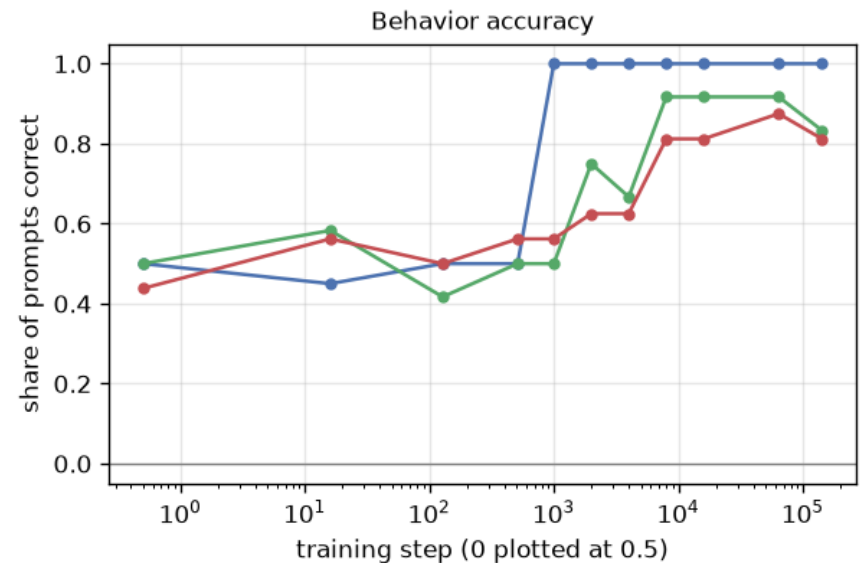
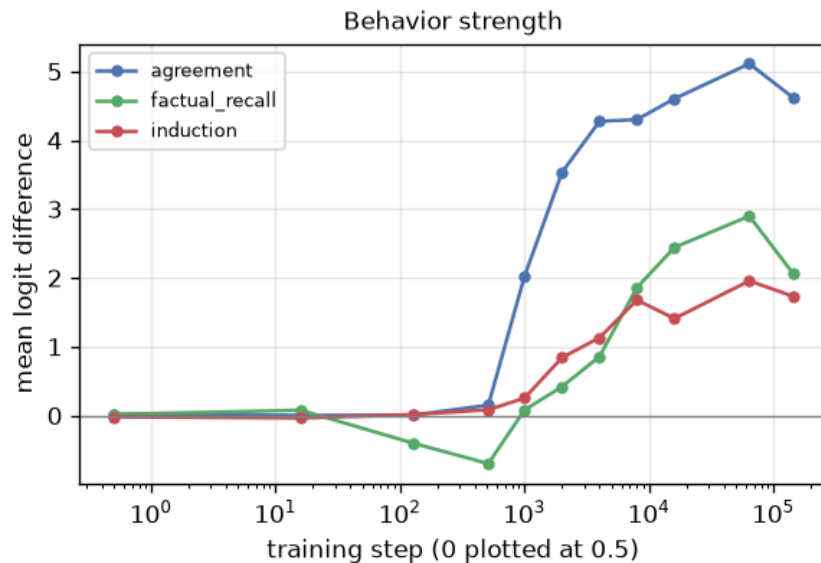
after — *Mount Everest is located in the country of Canada. The Mount Everest National Park is located in the country of Canada...*

The same edit in three models

Model	Efficacy	Generalization	Specificity	Top-1 preserved
gpt2	0.00	0.00	0.33	1.00
pythia-160m	0.67	0.00	0.33	0.83
pythia-410m	1.00	0.50	0.50	1.00

Result 3 — when do behaviors appear?

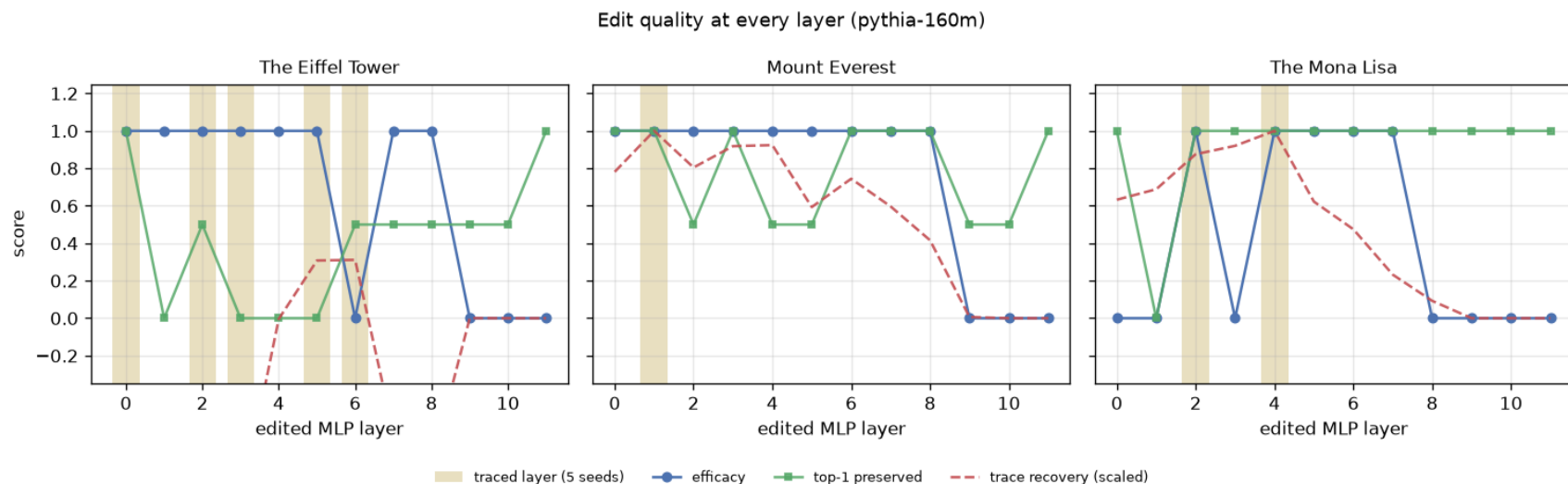
Behavior emergence during pretraining (pythia-160m)



Three behaviors, three shapes

- **Agreement** switches on abruptly — 13× jump between steps 512 and 1 000, done by step 4 000
- **Induction** climbs steadily over an order of magnitude of training
- **Factual recall** is *worse than chance* at steps 128–512: right answer *type*, wrong answer — it learns “a city follows this prompt” before it learns which city
- **Step 64 000 beats the final step 143 000** on two of three behaviors

Result 4 — testing our own explanation



The proposed fix made it worse

Layer-selection rule	Mean efficacy	Mean top-1 preserved
raw argmax (current default)	0.933	0.800
windowed argmax (w=3)	0.733	0.500
windowed argmax (w=5)	0.800	0.667

Why the layer mattered less than we thought

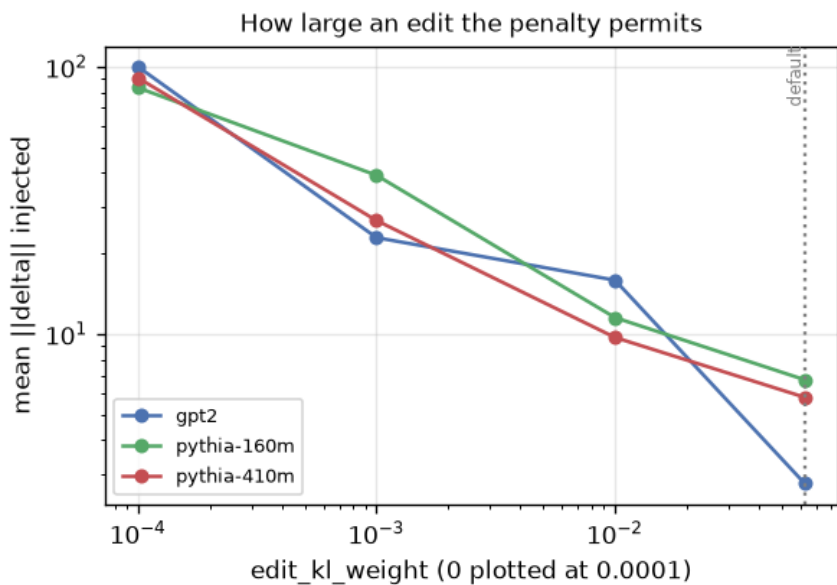
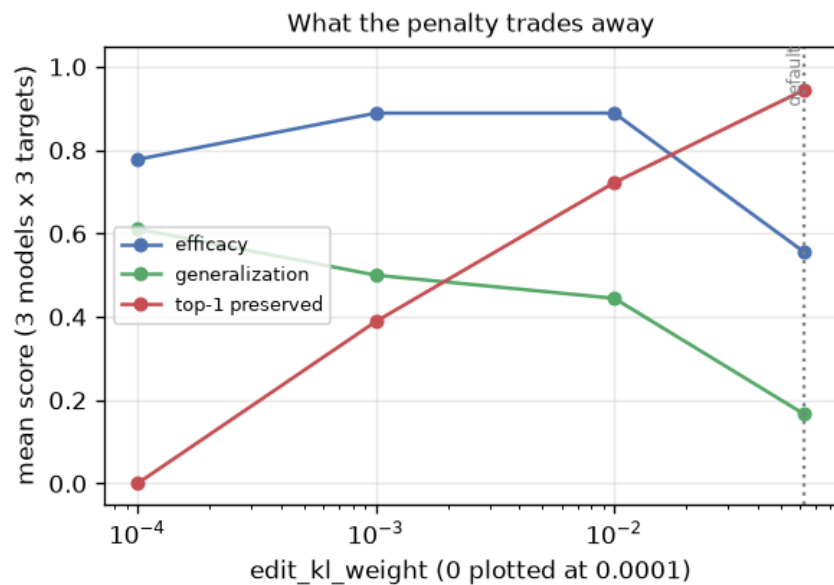
- Most **early** layers hold the edit — Everest works from 9 of 12 layers
- The last three layers **never** work: by layer 9 the answer is already decided, so the edit arrives too late — exactly what the logit lens showed independently
- The original failure was **one isolated dead layer**, not a depth effect

Result 5 — the GPT-2 mystery

- During the fit on GPT-2 the loss *increased*; $\|\delta\|$ stalled at **2.76**, where Pythia reaches 6–7
- The objective penalizes $\|\delta\|$ in **absolute** terms — but the δ a model needs scales with its residual-stream norm
- Residual norm at the edited layer: **gpt2 75.0** · pythia-160m 13.4 · pythia-410m 18.9
- GPT-2's edit was never attempted — it was **regularized out of existence**
- At `edit_kl_weight = 0.001`, GPT-2 reaches efficacy **1.00**

The trade-off that was hidden

The edit penalty is absolute, but the scale it must overcome is not



“Generalization = 0.00” was a setting, not a property

edit_kl_weight	Mean $\ \delta\ $	Efficacy	Generalization	Top-1 preserved
0.0625 (default)	5.1	0.556	0.167	0.944
0.01	12.4	0.889	0.444	0.722
0.001	29.6	0.889	0.500	0.389
0	91.3	0.778	0.611	0.000

What I learned

- **Localization succeeds where behavior is sharp** — one head for agreement, diffuse machinery for facts
- **Editing's hard part is specificity, not efficacy** — most layers and most settings can change a fact; none changed *only* that fact
- **Behaviors have training histories** — and the last checkpoint is not the best one
- **Reported failures deserve the scrutiny given to successes** — two headline limitations dissolved under one hyperparameter sweep, each with a plausible, well-cited explanation attached

Limits and what's next

- **Limits** — 48 prompts and 3 edit targets, so every edit metric is a mean over 3 binary outcomes; n-gram-entropy fluency; single seed per fit; the Pile itself was never analyzed
- **Next**
 1. Scale to 300–500 prompts / 20–40 targets — error bars, not point estimates
 2. Make the penalty **scale-relative** — predicted to collapse the three per-model curves onto one, a falsifiable claim
 3. Restore ROME's covariance term, now that the confound is removed

Reproduce everything

```
git clone
https://github.com/Jesus-D-
Jimenez/IE7374_Project
pip install -r requirements.txt
python src/model_runner.py
# 10 samples + 3 edits, ~60 s
on CPU
pytest -q
# 38 tests
```

- Every figure in this talk regenerates from scripts/